A Literacy-Free, Report-Free Eye Exam on a Consumer EEG Headband and VR Headset: Toward Objective Vision Screening by Community Health Workers

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***Abstract*—At least a billion people have a vision impairment that could be corrected, and the bottleneck is the screen: an eye test needs a literate, cooperating subject and a trained examiner. We describe an exam that asks nothing of the subject but to look at a dot with one eye covered, on a Muse headband with one electrode added over the visual cortex and a Meta Quest 3S running the stimulus in its browser, under 600 US dollars together. Acuity, contrast, field, gaze, colour and the apparent latency come from the steady-state visual evoked potential: an exact test per step, the sweep-VEP threshold rule and the slope of the response phase over flicker rate. A hardware-free phantom with set cut-offs and a dead quadrant passes all twenty-five checks of a full-stack gate. Three sittings on a person were null, and the rig's own records found why: the headset's browser had kept a cached page from before the flash test, so the flicker was logged but never shown. With the page verified, a fourth sitting gave a flash response in both eyes on the occipital electrode, each eye's best flicker rate and one acuity limb; it also showed that a contact rule written for loose electrodes had discarded a responding one, and that the latency fit is ambiguous in its turn count. Both are now method. Nothing here is a diagnosis.**

***Index Terms*—mobile health; visual screening; SSVEP; sweep VEP; wearable sensors; consumer electronics; community health workers.**

I. The Screen, Not the Lens, Is the Bottleneck

The WHO counts at least 2.2 billion people with a vision impairment, at least a billion of them addressable, uncorrected refractive error first among the causes [1]. A letter chart needs a subject who can read it in the examiner's language and an examiner to judge the answers. The pattern that has carried other screens to community health workers is a consumer device, a worker, and software that checks itself [2], [3]. The steady-state visual evoked potential (SSVEP) gives an objective acuity test with a clinical pedigree, the sweep VEP built for infants, who cannot report [4], and a contrast threshold [5]; it removes literacy, language and examiner bias while bringing biases of its own in hair and headband fit [6].

II. The Rig and the Exam

Mann et al. turned the eye into a camera with the SSVEP [7]; we rebuilt that rig on a Muse headband with a cup electrode at Oz, the phone streaming raw EEG as the hub in the wireless-sensing architecture iCalm set out [8], and a Quest 3S whose browser renders the stimulus (Figs. 1, 2).

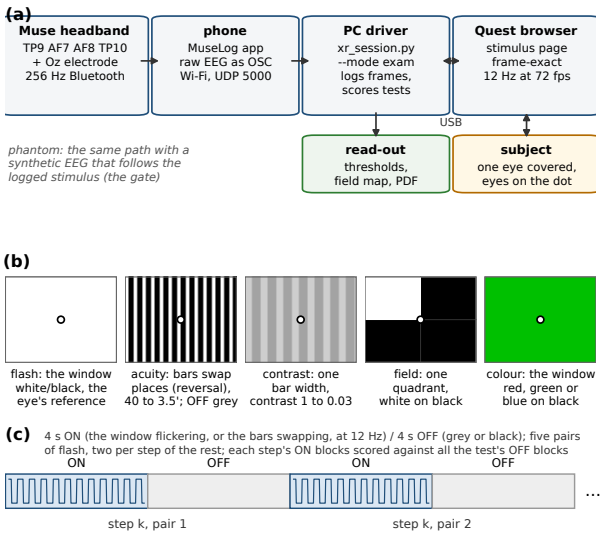


Fig. 1. (a) Scalp to read-out: the headband streams through the phone to the PC, which drives the page in the headset's browser, logs every frame and scores each test; the phantom replaces the first two boxes with a synthetic EEG that follows the logged stimulus. (b) The four stimuli and the OFF screen of the grating tests. (c) One step's blocks.

Every stimulus fills the window: the camera runs had shown the response grows with the flickering area and a small square gives nothing. Each eye opens with the window flickering white/black, the camera's calibration stimulus (five ON/OFF pairs of 4 s at 12 Hz on the 72 Hz display), which must respond and whose channel and amplitude are the reference for the eye's tests. The flash then runs at six rates, 6 to 36 Hz: amplitude against rate is the eye's temporal tuning, the slope of phase against rate its apparent latency [9], whose difference between the eyes is the clinical pattern VEP's measure. Then, two pairs per step: a grating whose bars swap places at the flicker rate, each resolved bar swinging like the flash at a constant mean luminance (pattern reversal [4]), 40 to 3.5 arcmin (acuity) or contrast 1 to 0.03 (contrast); one quadrant at a time (field); the whole window again while the dot sits at the centre and then 84% of the way to each edge, so the flicker falls to one side of the fovea (gaze); the window red, green or blue (colour). Each step's ON blocks are scored against all the test's OFF blocks by the line power at the

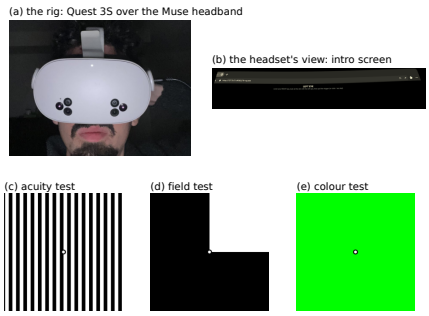


Fig. 2. (a) The Quest 3S over the Muse headband. (b) The headset's own view of one eye's intro screen. (c)–(e) The page during an ON frame of the acuity, field and colour tests.

flicker frequency and its double, with an exact permutation test over the channels in contact. Contact has two tiers: a channel above 300 μV rms through a test is open or railing and is set aside; between 60 and 300 μV it is noisy and kept only if its own ON/OFF test is significant. Thresholds follow the sweep-VEP rule: a line through the limb from the step below the finest responding one up the rise, extrapolated to zero; two asked widths the panel delivers as one are one step. The delivered stimulus is verified, not assumed: the page is served uncacheable at a fresh URL each sitting, its code is fingerprinted through the browser's debug port before the trigger, and the headset's own frame buffer is captured during a block. About nine minutes per eye.

III. Validation without a patient

A device for a worker with no eye clinic behind them must be testable with no patient in front of it. A phantom subject reads the stimulus log as it is written and streams synthetic EEG whose SSVEP follows what is on screen, through a logistic visibility in bar width and contrast with a cut-off per eye, dead quadrants, colour gains, and a latency and a flicker cut per eye. The gate dials in a right eye at 8 arcmin and contrast 0.08 with one dead quadrant, 120 ms late and holding to 40 Hz, a left eye twice as coarse and whole, 150 ms late and half gone by 24 Hz, gains 0.7/1/0.4 for red/green/blue, and asks twenty-five things: every stage delivered, the flash responding in both eyes on Oz and every test read on it, each threshold inside the band where the phantom's visibility runs from 0.02 to 0.6, the left eye at least 1.3× coarser, exactly the dead quadrant named, every gaze position responding, green $\geq$ red $>$ blue, every rate responding, the half rate and the latencies within 15% and 6 ms of the dialled. All twenty-five pass (Fig. 3): right eye 3.8 arcmin, contrast 0.029, quadrant 3 silent, 108 ms, still over half at 30 Hz; left eye 7.8 arcmin and 0.082, half rate 22.5 Hz, 30 ms later than the right; every gaze position 0.90 to 0.98 of the centre in both eyes. The gate earns its keep on every change: the first run with the gaze test failed exactly its two checks, because the phantom follows a fixed list of stimulus logs and the new one was not on it.

IV. Four sittings on a person

One of the authors ran the exam four times, right eye then left, at 12 Hz. The first three fitted nothing. The first

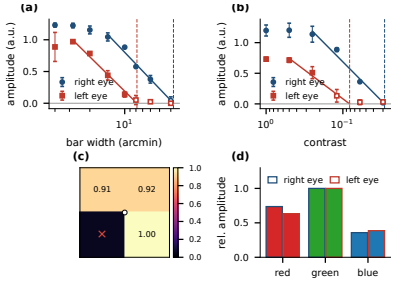


Fig. 3. The gate's read-out. (a) Acuity and (b) contrast per phantom eye (right circles, left squares; open: $p \geq 0.05$), fitted limbs and their zero crossings. (c) The right eye's field, the cross its dead quadrant. (d) Colour planes.

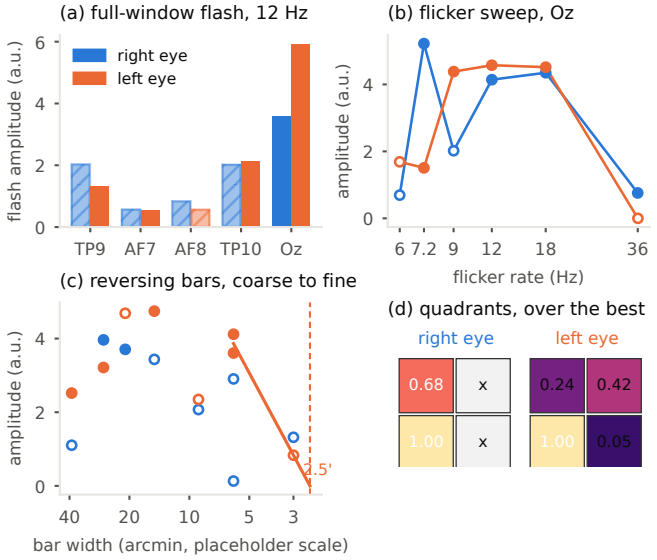


Fig. 4. The fourth sitting, both eyes on the occipital cup. (a) The flash per channel: right eye $p = 0.024$, left $p = 0.008$; hatched bars not significant. (b) The sweep: the right eye peaks at 7.2 Hz and is half gone by 8, the left peaks at 12 and holds to 25. (c) Reversing bars: the left eye responds to 6 arcmin and not to 3, the limb crossing at 2.5 on the headset's placeholder pixel scale; the right eye only to 21 and coarser. (d) Quadrants: the same pattern in both eyes, which points at the electrode, not the retinas.

taught the contact check: both ear electrodes had sat at 100 to 280 μV rms, loose under the headset strap, and the signal check, written for electrodes at the amplifier rails, had passed them, the quiet failure a consumer wearable brings to the field [3]; the exam now measures every channel before each eye and tells the subject which pad to press. The cause of all three surfaced in the fourth, in the headset's own frame buffer: during an ON block the view was black but for the dot, and the page's code, read through the browser's debug port, lacked the flash test. The browser had kept a cached copy of the page from a build before it, while the driver logged 72 frames per second of flicker from the page it thought it had. A logged stimulus is not a shown one; hence the delivery checks of Section II.

With the current page the fourth sitting measured (Fig. 4): a flash response on the occipital electrode in both eyes, right $p = 0.024$ at line SNR 9.6, left $p = 0.008$ at 18; a best rate of 7.2 Hz in the right eye and 12 Hz in the left, so the session rate suited one eye and not the other; bars resolved to

6 arcmin by the left eye, a limb extrapolating to 2.5 arcmin at the placeholder scale, against 21 arcmin by the right; no contrast limb in either; the same quadrant pattern in both eyes, bottom-left strongest and bottom-right weakest or silent, the signature of one electrode sitting off the midline between the two calcarine banks rather than of two retinas; and colour ratios that agree between the eyes (R:G:B 0.69:1:0.18 and 0.72:1:0.26), mostly the display's primaries, since a colour test proper exchanges two colours at equal luminance. Two methods changed on this data. The left eye had first been read on a forehead channel: its occipital electrode sat at 105 μV rms and the 60 μV rule set it aside, yet block by block it carried the sitting's strongest line, SNR 6 to 18 in every ON block against under 1.5 in every OFF block; hence the two tiers of Section II. And the apparent latencies, 77 ms right and 258 ms left, differ by a turn of phase, not by physiology: the phase step from 12 to 18 Hz is 209° in one eye and 197° in the other, and the fit took different turn counts because the eyes responded at different rate sets. The latency is reported as undetermined until the fit shares its turn count across the eyes, and the between-eye difference, the number the test exists for, needs both eyes responding at the same rates.

V. What the same recording carries

A headband records more than the visual response [10]. In companion sittings on the same rig the forehead electrodes gave the blink rate, which stops while a patch flickers, bursts when it goes dark, and doubled between two sittings of one evening, an old marker of fatigue [11]: the hook for a just-in-time adaptive intervention [12], an eye-strain estimate against the subject's own earlier minutes, not detection and not emotion.

VI. Toward the field

Established: the apparatus, an exact test per step, the sweep-VEP threshold rule, the latency from the phase slope, a phantom gate that catches failures without a volunteer, and on one person a flash response in both eyes, each eye's best rate and one acuity limb. Not established: a threshold in physical units, since the headset's pixels per degree is unmeasured; a latency free of its turn-count ambiguity; colour vision, which needs an isoluminant exchange rather than coloured flicker; and any reading of the quadrants until the electrode sits on the midline. Open after those: a blur study with trial lenses against a letter chart, a normative cohort, and a worker-facing flow with the report as one picture.

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